

COR-STAT1202 Introductory Statistics

Tutorial 3

Probability

Q1. Given that $X = \{1, 2, 4, 5\}$ and that $P(X = 1) = 0.2$, $P(X = 2) = 0.5$, $P(X = 4) = 0.2$, $P(X = 5) = 0.1$.

Calculate the following:

a) $P(X^2 \leq 10)$

b) $P(X > 1 \mid X \text{ is less than } 4)$

Answer:

a) The sample space of $X = \{1, 2, 4, 5\}$

X^2 less than or equal to 10: $\{1, 4\} \Rightarrow X = \{1, 2\}$

$$P(X^2 \leq 10) = P(X = 1) + P(X = 2) = 0.2 + 0.5 = 0.7$$

b) Let A be the event that $X > 1$: $A = \{2, 4, 5\}$

Let B be the event that X is less than 4: $B = \{1, 2\}$

$$A \cap B = \{2\} \Rightarrow P(X = 2) = 0.5$$

$$P(B) = P(X=1) + P(X=2) = 0.2 + 0.5 = 0.7$$

$$P(X > 1 \mid X \text{ is less than } 4) = P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.5}{0.7} = 0.7143$$

Q2. Mr Tan, a HDB approved contractor, has submitted two bids for two projects: Project X and Project Y. The probability of getting project X and Project Y is 0.60 and 0.75 respectively. The probability of him getting none of the projects is 0.15.

a) What is the probability that Mr Tan will get both projects?

b) Are the events of getting the two projects mutually exclusive? Why?

c) Are the events of getting the two projects independent? Why?

d) If Mr Tan gets project X, what is the probability that he will get project Y?

e) If Mr Tan gets project Y, what is the probability that he will get project X?

Answer:

a) $P(X) = 0.60$, $P(Y) = 0.75$, and $P(X \cup Y) = 1 - 0.15 = 0.85$

$$P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$$

$$\Rightarrow 0.85 = 0.60 + 0.75 - P(X \cap Y)$$

$$\text{Hence, } P(X \cap Y) = 0.50$$

b) Since $P(X \cap Y) = 0.50 \neq 0$, the two events are not mutually exclusive.

c) Since $P(X)P(Y) = (0.60)(0.75) = 0.45 \neq P(X \cap Y) = 0.50$, the events of getting the two projects are not independent.

d) $P(Y|X) = \frac{P(X \cap Y)}{P(X)} = \frac{0.5}{0.6} = 0.8333$

e) $P(X|Y) = \frac{P(X \cap Y)}{P(Y)} = \frac{0.5}{0.75} = 0.6667$

COR-STAT1202 Introductory Statistics

Tutorial 3

Probability

Q3. What is the probability of drawing a king or a spade from a pack of 52 poker cards?

Answer:

$$P(\text{king}) = \frac{4}{52} ; P(\text{spade}) = \frac{13}{52}$$

The events of drawing a king or a spade are not mutually exclusive, a king card can be a spade.

$$P(\text{king} \cup \text{spade}) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13} = 0.3077$$

Q4. Tiger Fund Ltd. operates a number of mutual funds in high technology and in financial sectors. Roberts is a fund manager who runs a major fund that includes a wide variety of technology stocks. As fund manager he decides which stocks should be purchased for the mutual fund. The compensation plan for fund managers includes a first-year bonus for each stock purchased by the manager that gains more than 10% in the first six months it is held.

Of those stocks that the company holds, 40% are up in value after being held for two years. In reviewing the performance of Roberts, they found that he received a first-year bonus for 60% of the stocks that he purchased that were up after two years. He also received a first-year bonus for 40% of the stocks he purchased that were not up after two years.

What is the probability that a stock will be up after two years given that Roberts received a first-year bonus?

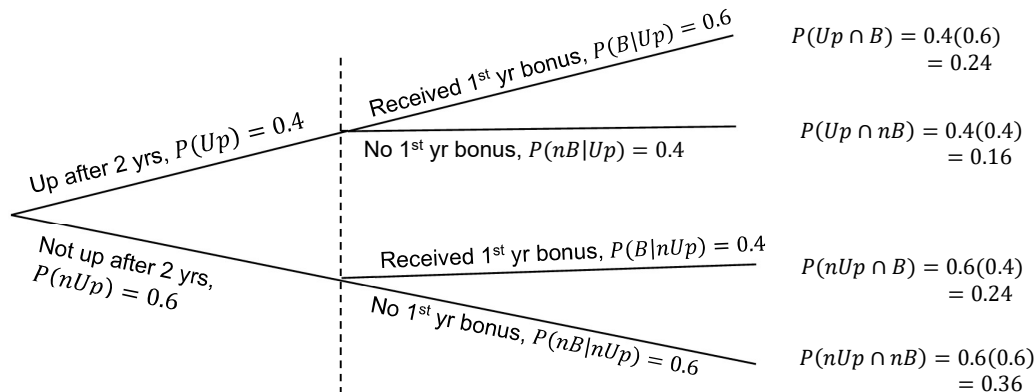
Answer:

Let Up : stocks that are up after being held for 2 years

nUp : stocks that are not up after being held for 2 years

B : received first-year bonus

nB : no first-year bonus



$$\begin{aligned} P(Up|B) &= \frac{P(B|Up)P(Up)}{P(B)} = \frac{P(Up \cap B)}{P(Up \cap B) + P(nUp \cap B)} \\ &= \frac{0.6(0.4)}{0.4(0.6) + 0.6(0.4)} \\ &= 0.5 \end{aligned}$$

COR-STAT1202 Introductory Statistics

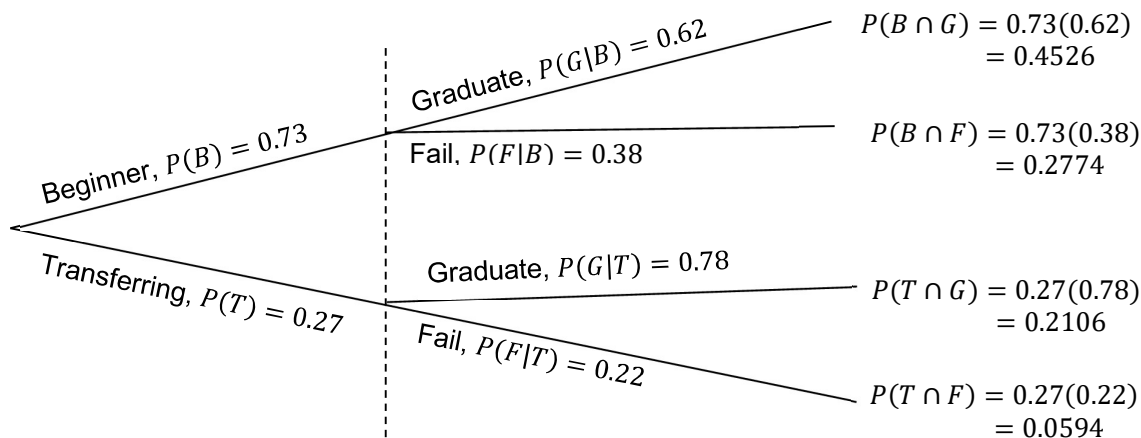
Tutorial 3

Probability

Q5. A student can enter a course either as a beginner (73%) or as a transferring student (27%). It is found that 62% of beginners eventually graduate, and that 78% of transfers eventually graduate. Find:

- the probability that a randomly chosen student is a beginner and eventually graduate.
- the probability that a randomly chosen student will eventually graduate.
- the probability that a randomly chosen student is either a beginner or will eventually graduate, or both.
- If a student eventually graduates, what is the probability that the student entered as a transferring student?
- If two entering students are chosen at random, what is the probability that not only do they enter in the same way but that they also both graduate or fail?

Answer:



- $P(B \cap G) = 0.4526$
- $P(G) = P(B \cap G) + P(T \cap G) = 0.6632$
- $P(\text{Beginner or eventually Graduate or both}) = 1 - P(F \cap T) = 1 - 0.0594 = 0.9406$
Or $P(\text{Beginner or eventually Graduate or both}) = P(B \cap G) + P(B \cap F) + P(T \cap G) = 0.4526 + 0.2774 + 0.2106 = 0.9406$
- $P(T|G) = \frac{P(T \cap G)}{P(G)} = \frac{0.2106}{0.6632} = 0.3176$
- Both students are transfers and graduate: $P(T \cap G) = 0.2106(0.2106) = 0.0444$
Both students are transfers and fail: $P(T \cap F) = 0.0594(0.0594) = 0.0035$
Both students are beginners and graduate: $P(B \cap G) = 0.4526(0.4526) = 0.2048$
Both students are beginners and fail: $P(B \cap F) = 0.2774(0.2774) = 0.0770$
Probability of both students enter in the same way but that they also both graduate or fail
 $= 0.0444 + 0.0035 + 0.2048 + 0.0770 = 0.3297$

COR-STAT1202 Introductory Statistics

Tutorial 3

Probability

Extra Questions

1. In an experiment, three dice were thrown into five cylinders at random. Assuming each die has the same chance of going into any cylinder independently of where the other 2 dice fall. Determine the probability of 0, 1, 2, 3, 4, and 5 empty cylinders.

Answer:

The situations below will never happen since there are three dice:

$$P(0 \text{ cylinders empty}) = 0$$

$$P(1 \text{ cylinders is empty}) = 0$$

The situations below will never happen since all dice has to be in at least of the cylinders:

$$P(5 \text{ cylinders empty}) = 0$$

$P(2 \text{ cylinders empty}) = P(1^{\text{st}} \text{ dice occupying any cylinders and } 2^{\text{nd}} \text{ dice occupying a cylinder that is empty and the } 3^{\text{rd}} \text{ dice occupying a cylinders that is empty})$

$$= \frac{5}{5} \left(\frac{4}{5} \right) \left(\frac{3}{5} \right) = \frac{12}{25}$$

$P(4 \text{ cylinders empty}) = P(1^{\text{st}} \text{ dice occupying any cylinders and } 2^{\text{nd}} \text{ dice occupying the same cylinders as the } 1^{\text{st}} \text{ dice and } 3^{\text{rd}} \text{ dice occupying the same cylinders as the } 1^{\text{st}} \text{ dice})$

$$= \frac{5}{5} \left(\frac{1}{5} \right) \left(\frac{1}{5} \right) = \frac{1}{25}$$

$P(3 \text{ cylinders empty})$

$$= 1 - P(1 \text{ cylinders empty}) - P(2 \text{ cylinders empty}) - P(4 \text{ cylinders empty}) - P(5 \text{ cylinders empty})$$

$$= 1 - 0 - \frac{12}{25} - \frac{1}{25} - 0$$

$$= \frac{12}{25}$$

Alternatively,

$1^{\text{st}} \text{ dice} - \text{can go into any cylinder, } P(1^{\text{st}} \text{ dice in a cylinder}) = \frac{5}{5} = 1$

$2^{\text{nd}} \text{ dice} - \text{same cylinder as } 1^{\text{st}} \text{ dice, } P(2^{\text{nd}} \text{ dice in the same cylinder as } 1^{\text{st}} \text{ dice}) = \frac{1}{5}$

$3^{\text{rd}} \text{ dice} - \text{any other cylinder that is empty, } P(3^{\text{rd}} \text{ dice in any of the 4 empty cylinders}) = \frac{4}{5}$

OR

$2^{\text{nd}} \text{ dice} - \text{not the same cylinder as the } 1^{\text{st}} \text{ dice, } P(2^{\text{nd}} \text{ dice in any of the 4 empty cylinders}) = \frac{4}{5}$

$3^{\text{rd}} \text{ dice} - \text{same cylinder as either } 1^{\text{st}} \text{ or } 2^{\text{nd}} \text{ dice, } P(3^{\text{rd}} \text{ dice in same cylinder as } 1^{\text{st}} \text{ or } 2^{\text{nd}} \text{ dice}) = \frac{2}{5}$

$$\text{Total probability} = 1 \left(\frac{1}{5} \right) \left(\frac{4}{5} \right) + 1 \left(\frac{4}{5} \right) \left(\frac{2}{5} \right) = \frac{12}{25}$$

COR-STAT1202 Introductory Statistics

Tutorial 3

Probability

2. a) What is the probability of at least one odd number if two fair dice are thrown?
b) What is the probability of at least one even number if three fair dice are thrown?

Answer:

- a) Total possible number of outcomes with throw of two dices = 36
Total outcomes in event with at least one odd number = $36 - 9 = 27$ (there are only 9 pairs of both even numbers)

$$P(\text{at least one odd number}) = \frac{27}{36} = 0.75$$

- b) Total possible number of outcomes with throw of three dices = $6 \times 6 \times 6 = 216$
Number of outcomes with all odd numbers: $3 \times 3 \times 3 = 27$

Number of outcomes with at least one even number = $216 - 27 = 189$

$$P(\text{at least one even number}) = \frac{189}{216} = 0.875$$

3. A fair dice is thrown twice. Determine the following:

- a) A one turns up each time
b) The sum of the scores is 9
c) Both are odd

Answer:

- a) Total possible number of outcomes with throw of two dices = 36
Event with a one turns up each time: $\{(1,1)\}$

$$P(\text{a one turns up each time}) = \frac{1}{36}$$

- b) Total possible number of outcomes with throw of two dices = 36
Event with sum of the scores is 9: $\{(3,6), (4,5), (6,3), (5,4)\}$

$$P(\text{sum of the scores is 9}) = \frac{4}{36} = \frac{1}{9}$$

- c) Total possible number of outcomes with throw of two dices = 36
Event with both odd: $\{(1,1), (1,3), (1,5), (3,1), (3,3), (3,5), (5,1), (5,3), (5,5)\}$

$$P(\text{sum of the scores is 9}) = \frac{9}{36} = \frac{1}{4}$$

4. The number of male and female students taking different modules are shown in the table below.

Module	Male (M)	Female (F)
Statistics (S)	5	14
Economics (E)	17	41
Accounting (A)	4	25
Programming (P)	23	11
Total	49	91

- a) Discuss and justify if the event of taking Statistics and being female are independent events.
b) Given that a student takes Economics, what is the probability of this student is a male.

COR-STAT1202 Introductory Statistics

Tutorial 3

Probability

Answer:

a) Since $P(S) = \frac{19}{140} = 0.1357$, $P(F) = \frac{91}{140}$

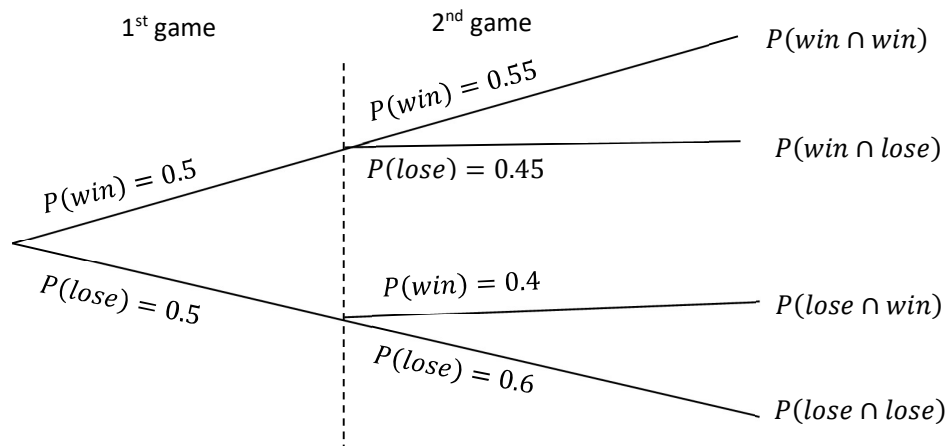
$$P(S)P(F) = \frac{19}{140} \left(\frac{91}{140} \right) = 0.0882 \neq P(S \cap F) = \frac{14}{140} = 0.1 \Rightarrow S \text{ (taking Statistics) and } F \text{ (being Female) are not independent events.}$$

b) $P(M|E) = P(M \cap E) = \frac{P(M \cap E)}{P(E)} = \frac{17}{58} = 0.2931$

5. The badminton player, Loh Kean Yew, plays two games against an opponent. He has a 50% chance of winning the first game, and if he wins the first, he then has a 55% chance of winning the second game. But if he loses the first, he has only a 40% chance of winning the second game.

- a) What is the probability of him winning both games?
b) If he lost the second game, what is the probability he lost both games?

Answer:



a) Probability of winning both games, $P(\text{win} \cap \text{win}) = 0.5 \times 0.55 = 0.275$

b) $P(\text{lost both games} | \text{lost 2nd game})$

$$= \frac{P(\text{lost 1st game} \cap \text{lost 2nd game})}{P(\text{lost 2nd game})}$$

$$= \frac{0.5 \times 0.6}{(0.5 \times 0.45) + (0.5 \times 0.6)}$$

$$= 0.5714$$